

1. Save the following data table in a file :

x	2.0	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
$f(x)$	-1.118	0.074	1.508	3.06	4.572	5.862	6.75	7.07	6.7	5.576	3.709

Fit the data points of the second row against that of the first row using GNUPLOT with the equation $f(x) = x^a \sin bx$. Write the values of the fitted parameters a , b . Show the fitted curve along with the data points in the same plot. Set the title, x-label, y-label as 'Data fitting', ' x ' and ' $f(x)$ ' respectively. Save the plot in suitable format. 20

2. Define the following function $f(x) = \begin{cases} \frac{4}{x^2 + 1} & x \leq 1 \\ x^2 + 1 & 1 < x < 3 \\ 2x + 4 & x \geq 3 \end{cases}$

Set the range of x values from -1 to $+4$ and the range of y values from -1 to $+12$. Show both the axes on the plot. Choose suitable sample size to make the functions actually continuous in the graph. Set title "Continuous function with discontinuous first derivative". Plot the function, store it in a file and print it. 20

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3. Plot the following polar functions for $-\pi \leq \theta \leq \pi$.

$$r(\theta) = 2\sqrt{\cos \theta}, \text{ and } r(\theta) = -2\sqrt{\cos \theta}$$

Set different line styles and line widths for these two plots. Put the title, x-label and y-label as 'Polar plots', ' θ ' and ' $r(\theta)$ ' respectively. Set sample 800. Save the plot in suitable format. 20

4. Plot the functions $(\sin x)e^{-x}$, $x \cos x$ and $x^2 \sin x$ with the same $x-y$ axes. Put the title, x-label and y-label as 'Function plots', ' x ' and ' $f(x)$ ' respectively. Set different line styles and line widths for different plots. Set samples to 800. Save the plot in suitable format. 20

5. Plot the following parametric functions :

$$x(t) = e^{\frac{1}{2}} \cos\left(\frac{\pi}{4}t\right) + \cos(2\pi t), \quad y(t) = \frac{e^{\frac{3}{2}}}{2} \sin(\pi t) + \sin(2\pi t),$$

for $-1 \leq t \leq 1$. Put the title as 'Parametric plot'. Set samples to 800. Save the plot in suitable format. 20

6. Make the following data table in a file.

x	1.5	2.81	3.19	5.32	7.24	11.79
y	19.14	18.40	18.18	16.90	15.87	13.24

Plot the data in second column against that of the first column. Set the range of x values from 0 to 15. Try to fit the data with the equation of type $f(x) = mx + c$. Show the points and the fitted line on the same plot. Set suitable title, x-label and y-label. Mention the values of m and c after fitting the data. Store the plot in a file and print it. 20

(3)

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7. Define three functions in the given range

$$f_1(x) = \frac{1}{x} \quad x > 0.5$$

$$f_2(x) = \frac{1}{x^2 + 1} \quad -0.5 < x < 0.5$$

$$f_3(x) = x^2 \quad x < -0.5$$

Set the range of x values from -1.5 to $+1.5$ and the range of y values from 0 to + 3. Show y axis on the plot. Set title 'Piecewise continuous function'. Set ticks of the y axis at an interval of 0.5 on the axis. Plot the three functions and store the plot in a file and print it. 20

Group - B

8. (a) Using the Newton-Raphson method with starting point $x_0 = 3$, determine the root of the following equation correct up to three decimal places :

$$x^3 + 5x^2 - 19x + 3 = 0.$$

Print the root.

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- (b) Write a Python program to calculate the value of $\cos \frac{2\pi}{7}$ using the following infinite series :

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

correct up to three decimal places. Print the value.

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9. (a) Find a real root in between $[0, 3]$ of the following equation $3x^3 - 5x + 1 = 0$ correct up to three decimal place using Bisection method. Store the root and the value of the function at the root and print it.

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- (b) Find the minimum number of terms n required to make the sum of the following series S greater than $m = 10$.

$$S = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} +$$

Take m as runtime input. Also print the value of the sum for that number of terms.

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10. (a) The velocity of a particle moving in 1D is given by $\frac{dx}{dt} = v_0 e^{-at}$, where $v_0 = 2\text{m/s}$, $a = 0.2/\text{s}$ and $x(0) = 0.0$. Using Euler method, solve the above equation to find the displacement x at time $t = 10\text{s}$, choose the step size to be less than 0.2. Store the t and x data in a file. No need to print the data file.

Given that the exact solution is $v(t) = \frac{v_0}{a} (1 - e^{-at})$. Compare the calculated and exact values at $t = 10\text{s}$. Print both the values.

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- (b) Create a list of all prime numbers between 10 to 100. Print the list.

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11. (a) Using Newton-Raphson method with starting point $x_0 = 2.5$, determine the root of the following equation correct up to three decimal places :

$$5x \sin x - 3x = 0$$

Print the root.

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- (b) Arrange the following numbers in ascending order using bubble or selection method :

$$2.91, 0.48, -2.76, -1.42, 1.63$$

Print the arranged numbers.

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12. (a) A first-order differential equation is given as below :

$$\frac{dy}{dx} = y^2 e^x \text{ with } y(0) = -1$$

Create a list (say X) with 100 x -values between $[0, 1]$. Write a Python program to solve this differential equation using Euler method for x -values in the list X . It is given that the exact solution is $y(x) = -e^{-x}$. Compare the calculated and exact values at $x = 1$. Print both the values. 30

- (b) Calculate the roots of the quadratic equation $x^2 + x - 6 = 0$. Print the solutions. 10

13. (a) Write a Python program to find the root of the transcendental equation correct up to three decimal places

$$x^2 + e^x \sin x = 0$$

in between $[2, 4]$ using Bisection method. Print the root.

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- (b) Two numbers are given as $F_0 = 0$ and $F_1 = 1$. Later numbers are defined as $F_n = F_{n-1} + F_{n-2}$. Create a list of numbers $F_0, F_1, F_2, \dots, F_{10}$. Print the list. 10

14. (a) Write a Python program to verify the following matrix identity

$$(AB)^T = B^T A^T$$

for

$$A = \begin{pmatrix} -3 & 1 & -2 \\ 1 & 0 & 4 \\ 2 & -4 & -2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 4 & -1 & -2 \\ -3 & 5 & 0 \\ -2 & -4 & 2 \end{pmatrix}$$

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- (b) Do the sum (S) of the following series :

$$S = -10, -9, \dots, 0, 1, 2, 3, \dots, 9, 10$$

Print S .

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